Local–Global Information Perception Network for Salient Object Detection in Optical Remote Sensing Images

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***Abstract*— In the field of salient object detection (SOD), optical remote sensing images (ORSI) differ significantly from natural scene images (NSI). The existing research in ORSI-based SOD is constrained by the limitations of convolutional neural networks (CNNs) in feature extraction and by the underutilization of feature information in Transformer-based approaches. To address these challenges, this article presents a Transformer-based local–global information perception network (LGIPNet) for ORSI, which enhances encoder-generated features at multiple levels to highlight salient targets through three specialized feature enhancement modules. The edge adaptive enhancement module (EAEM) focuses on extracting local edge features to guide precise edge generation. The multiscale grouped weighted attention module (MGWAM) extracts the local information from low-level features, scales features, and uses multiscale channel and learnable weighted spatial attention to locate salient targets. For high-level features, the dual-domain attention module (DDAM) integrates a channel enhancement attention (CEA) block and an adaptive spatial attention (ASA) block to refine both local and global information. Specifically, the EAEM sharpens the edges of salient objects, thereby ensuring the precision of boundary detection. The MGWAM, on the other hand, enriches the feature representation across multiple scales, enhancing the network's capability to encapsulate both fine-grained details and broader contextual information. The DDAM further strengthens the balance between local and global information, preserving feature integrity across levels. Finally, multiscale features are cascaded to produce the final saliency map. This holistic strategy empowers LGIPNet to accurately identify and emphasize salient objects in ORSI. Experiments on three datasets demonstrate that LGIPNet outperforms existing state-of-the-art methods, establishing its effectiveness and robustness in ORSI-based SOD. The source code is available at: https://github.com/sCauliflower/LGIPNet.git**

***Index Terms*— Dual-domain attention module (DDAM), edge adaptive enhancement module (EAEM), multiscale grouped weighted attention module (MGWAM), salient object detection (SOD).**

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I. Introduction

SALIENT object detection (SOD) is a critical task in computer vision (CV), aimed at replicating the human visual system's ability to accurately segment visually prominent areas within images [1], [2], [3]. SOD improves processing efficiency by automatically locating and segmenting salient regions, thereby providing essential support for subsequent advanced visual tasks. Serving as an exemplary preprocessing step, SOD has garnered noteworthy accomplishments in diverse CV domains, including but not limited to target tracking [4], image classification [5], [6], [7], image fusion [8], and image quality assessment [9], [10], [11]. In optical remote sensing images (ORSIs), SOD can accurately identify and highlight objects of interest, such as special features and key facilities. The ORSI-SOD has been widely adopted in diverse disciplines, including agriculture, military, and ecological conservation [12], [13].

ORSI and natural scene images (NSI) differ significantly in their imaging characteristics, orientation, proportion, and content [14], [15]. These distinctions lead to variations in resolution characteristics, target typologies, and scale distributions. Consequently, NSI-SOD methods are often challenging to apply directly to ORSIs. ORSI-SOD encounters specific difficulties, including low contrast in most remote sensing targets, complex backgrounds, and the coexistence of targets at various scales. These intrinsic characteristics necessitate dedicated architectural designs to enhance models' semantic comprehension capabilities for ORSI-specific features.

In recent years, both convolutional neural network (CNN)-based and Transformer-based methods have achieved significant progress in ORSI-SOD. Li et al. [16] proposed a method, which employed L-shaped and V-shaped modules to capture local details and handle complex backgrounds. However, this method overlooks the interaction between features

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of different dimensions, resulting in incomplete detection results for blurred targets. Zhang et al. [17] constructed an approach to enhance deep features through a multilevel attention flow structure. Nevertheless, this approach has insufficient processing of edge details, often leading to blurred edge detection. Li et al. [18] proposed a scheme to integrate multilevel features to highlight salient regions. However, the encoder based on CNN has difficulty capturing long-range context dependencies, making it prone to misdetections in complex backgrounds. Unlike CNNs, Transformers excel at capturing global contextual information, demonstrating superior performance in handling large-scale targets and complex backgrounds. Wang et al. [19] proposed a hybrid feature alignment strategy, which integrated multilevel feature fusion with Transformer mechanisms to enhance long-range contextual modeling capabilities. By establishing global and local information interactions across different feature levels, their method improves the accuracy of SOD, particularly by reducing false positives and false negatives in complex background scenarios.

Although existing methods have achieved remarkable accuracy in ORSI-SOD, there are still many limitations. First, most methods rely on CNN to encode the features of ORSI. However, CNNs possess a limited receptive field, making it difficult to model long-range spatial dependencies within ORSIs. Consequently, detection can be incomplete, and salient objects are prone to being missed. Second, methods, such as EMFINet [20] and MJRBM [21], use edge ground-truth (GT) supervision. These methods assume that the edge information is independent and complete. However, in ORSI, edges are affected by factors such as noise, and the GT labels may not accurately represent the diverse and incomplete boundaries of the targets. This may lead to the phenomenon of overfitting and limit the generalization ability of the model. Third, existing methods often over-rely on conventional channel and spatial attention mechanisms. The channel attention mechanism may compress crucial information when processing deep features, potentially leading to irrecoverable information loss. The spatial attention mechanism uses pooling operations and fixed-size convolutions, making it difficult to generate flexible attention maps and unable to fully capture the structural features of ORSI, thus risking the omission of salient objects.

In view of the above problems, we propose the local–global information perception network (LGIPNet). This architecture employs the pyramid vision Transformer (PVT) [22] backbone network to explore multiscale features and long-range spatial dependencies, thus overcoming the limitations inherent in CNN-based approaches. To address the issue of edge blurring, the edge adaptive enhancement module (EAEM) is introduced. This module uses feature grouping attention and differential norm pooling blocks (DNPBs) to adaptively enhance edge features without relying on GT labels. We designed the multiscale grouped weighted attention module (MGWAM) and the dual-domain attention module (DDAM) to extract rich semantic information in the deep features. MGWAM aims to overcome the limitations of existing methods in channel information compression loss and the insufficient adaptability of spatial attention mechanisms. It uses multiscale feature sampling combined with dilated convolutions to strengthen semantic feature extraction and enhance feature dependencies through channel attention. At the same time, a learnable grouped weighted spatial attention mechanism is introduced to dynamically adjust the weights of the target area, ensuring the integrity and consistency of salient objects. Through a cascade and weighted fusion strategy, MGWAM effectively suppresses complex background interference and preserves key feature details. DDAM consists of the channel enhancement attention (CEA) and the adaptive spatial attention (ASA) blocks. Specifically, CEA uses dual-frequency global pooling to retain key features and reduce the loss of small-object information. Simultaneously, ASA integrates multiscale depthwise separable convolutions to adaptively adjust the receptive field and enhance the discriminability of small objects in complex backgrounds. Furthermore, DDAM adopts a joint channel–spatial weighted strategy to further improve the integrity and saliency of small objects.

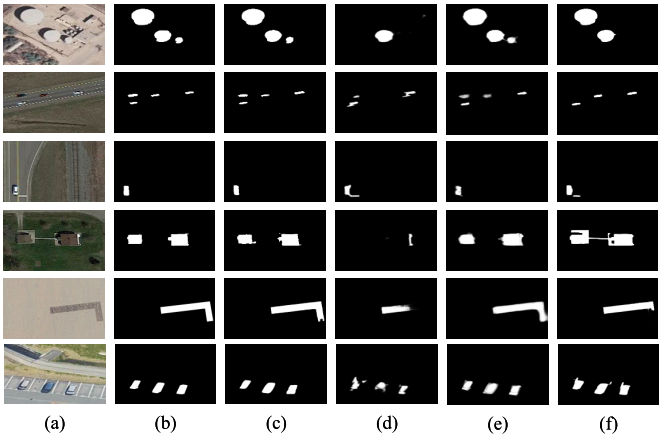


Fig. 1. Performance comparison of advanced SOD methods and our method in terms of visualization results. (a) ORSI. (b) GT. (c) Proposed LGIPNet. (d) LVNet [16]. (e) DAFNet [17]. (f) MCCNet [18].

Combining these techniques, our proposed LGIPNet can generate saliency maps with complete objects and clear edges. In Fig. 1, we showcase the saliency maps generated by our method compared to those from previous detection methods. As illustrated in Fig. 1(c), this demonstration not only confirms the effectiveness of the introduced modules but also indicates that our method outperforms other approaches in terms of detection results.

The contributions can be summarized as follows.

1) We construct a novel LGIPNet, employing a Transformer backbone network as an encoder and carefully considering semantic information at various levels. This architecture effectively addresses challenges in ORSI-SOD, such as detection redundancy, blurred edge detection, and the omission of small-target amidst complex backgrounds.
2) The proposed EAEM adopts a feature grouping strategy and attention mechanism to enhance the target feature representation. It utilizes DNPBs to adaptively enhance edge features within and across these feature groups, thereby improving the network's edge detection ability.

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3) The proposed MGWAM integrates a multiscale strategy with group-weighted spatial attention. It utilizes a modified channel attention to enhance feature dependencies across multiple scales and employs a learnable group-weighted spatial attention to aggregate the spatial information of targets. This module accurately locates the positions and contours of salient targets, thus refining the overall detection performance.
4) The proposed DDAM is designed to enhance the interaction between features at different levels by combining CEA and ASA. CEA employs dual-frequency global pooling to mitigate the loss of global information. Then, ASA perceives deeper spatial information across different receptive fields, thereby enriching the expression of small-scale features and improving the overall feature representation.

II. Related Works

ORSI-SOD has developed rapidly in recent years. In this section, we review some traditional and CNN-based methods of ORSI-SOD and describe the application and development of Transformer in the visual field.

A. SOD in ORSI

ORSI-SOD plays a crucial role in understanding ORSIs. In recent years, numerous ORSI-SOD methods have been proposed, among which CNN-based approaches have dominated the field and achieved remarkable performance. This section focuses on traditional and CNN-based methods.

The development of ORSI-SOD is similar to the development process of natural image saliency detection. Early researchers tried to achieve this by analyzing traditional methods such as pixel characteristics of ORSIs. Faur et al. [23] proposed a saliency segmentation method based on rate-distortion measurement. Zhang et al. [24] proposed a saliency detection model based on the analysis of color information content, which applied the color information content of optical RSIs and outputs a full-resolution saliency map. Zhang et al. [25] proposed a detection model based on the combination of superpixel and statistical saliency feature analysis. This model introduced a statistical saliency feature map through the histogram saliency analysis in the Laboratory color space. Then, based on the combination of superpixel segmentation and information entropy, an information saliency feature map was obtained. Traditional ORSI-SOD has limitations in information extraction, and many traditional methods require a lot of preliminary manual work, which makes detection accuracy and efficiency unsatisfactory and has not attracted continuous attention from researchers.

However, with the advantages of CNN in the field of CV and its powerful ability to extract image features, many researchers have turned their attention to this field and proposed excellent solutions. Huang et al. [26] explored the intrinsic properties of multilayer features. By aggregating multiple high-level features, they roughly but accurately located multiscale objects, and then guided the integration of subsequent features in a step-by-step feedback manner, making full use of multilevel features. Zhou et al. [27] paid more attention to edge details and feature fusion. They formed a U-shaped architecture through the edge extraction module and encoder to obtain accurate edge information and sharpened the edge information through attention, proposing corresponding solutions for difficult edge detection. Li et al. [28] proposed an adjacent context coordination module, which contained a local branch and two adjacent branches, to coordinate multilevel features at the same time. The local branch adaptively highlighted the salient areas, while the adjacent branches introduced global information of adjacent levels to enhance the salient areas. Sun et al. [29] designed a channel enhancement module to enhance and interact features of various dimensions, and generated global attention features through dimension transformation and feature relationship aggregation, thus achieving the function of locating salient targets. Lee et al. [30] proposed a network leveraging structural priors and hierarchical feature updates to enhance SOD in complex backgrounds. The method integrates structural prior information with a hierarchical feature update strategy, guiding the feature extraction process and progressively refining salient object representations, thereby improving detection accuracy. Although those methods from local to global have made significant progress, the current research on context interaction mainly relies on convolutional modules to explore the interaction between features of different levels.

B. Transformer for SOD

The Transformer model was initially applied in the field of natural language processing. However, due to its powerful representational ability, researchers began to explore its application in CV tasks. The main advantage of the Transformer model lies in its self-attention mechanism, which allows the model to capture long-distance dependencies between elements in the input sequence and supports parallel processing of sequences, contrasting with recurrent networks (such as long short-term memory networks). Unlike CNNs, the Transformer has minimal inductive bias in its design and is naturally suited as a set function. In addition, the direct design of the Transformer allows the use of similar processing blocks to handle multiple modalities, and it has excellent scalability for very large networks and huge datasets [31], [32], [33].

Researchers have begun to explore its application in CV tasks, including salient detection. Wang et al. [34] proposed a new Transformer framework for saliency detection, named TF-SOD, the framework mainly consisted of the encoder part of the fully convolutional network, fusion module, Transformer module, and feature decoding module. This architecture allowed the network to achieve a tight integration of local and nonlocal interactions, complementing information and deeply mining the correlation information between features. Liu et al. [35] proposed a new model based on the Transformer, named visual saliency Transformer (VST), for RGB and RGB-D SOD. It took image blocks as input and used the Transformer to propagate global context between these blocks. Liu et al. [36] proposed a cross-modal fusion model based on Swin Transformer, named SwinNet, for RGB-D and RGB-T SOD.

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It extracted hierarchical features, bridges the gap between two modalities through the attention mechanism, and was guided by edge information to sharpen the contours of salient objects. Ren et al. [37] designed a simple and effective deep transformation decoder based on the Transformer encoder. The decoder densely decoded and upsampled Transformer features to generate the final saliency map with less injected noise. Chen et al. [38] proposed a collaborative compensative Transformer network, featuring a collaborative relation Transformer for enhanced multiscale contextual modeling and a compensative fusion Transformer for efficient cross-level feature interaction. By incorporating a feature compensation mechanism, the method reduces object-background confusion and improves the boundary integrity, effectively addressing saliency detection challenges in complex backgrounds.

Recently, scholars have adopted the Transformer scheme to solve ORSI-SOD. Bai et al. [39] proposed a global–local–global context-aware network, which employed a Transformer-based encoder to extract the global representation and combined local context-aware features collected from three saliency enhancement modules for integrated saliency modeling, as well as an edge assignment module to optimize the initial detection. Dong et al. [40] proposed a Transformer-based framework with a large-kernel convolutional decoder to enhance feature modeling and edge preservation, effectively capturing long-range dependencies and improving small-object and background discrimination. Zhao et al. [41] proposed an adaptive dual-stream sparse Transformer network that integrated Transformer's global modeling with CNN's local detail extraction to enhance multiscale SOD. It employs a dual-stream architecture, where the Transformer captures long-range dependencies, CNN refines edges and details, and sparse attention reduces computational cost while improving detection performance, offering new insights for ORSI-SOD. In summary, researchers are trying to investigate Transformer-based methods or CNN and Transformer fusion-based methods to cope with ORSI-SOD more effectively.

However, the existing research paradigm mainly focuses on the extraction and exploitation of generalized information, while the in-depth application of specific features is still insufficient. Current approaches usually enhance the detection effect by simply extracting multiscale features but lack the fine modeling and effective use of key features, such as edge features and texture features. Due to the complexity and diversity of remote sensing images, this kind of rough feature processing may limit the sensitivity and accuracy of the model to salient regions. Therefore, it is necessary to further refine the feature processing strategy on the basis of feature extraction to better adapt to the requirements of high accuracy and high resolution of the ORSI-SOD.

III. Proposed Method

In this section, we provide a detailed overview of the overall structure of the proposed LGIPNet. The framework consists of three main components: an encoder based on feature extraction, three modules for feature enhancement (i.e., EAEM, MGWAM, and DDAM), and a decoder for saliency map generation.

A. Network Overview

In Fig. 2, we illustrate the overall architecture of our proposed LGIPNet, which adopts an encoder network, three feature enhancement processing modules, and a decoder to form an end-to-end encoder-decoder structure. We employ the popular Transformer-based PVT [42] network as the encoder of the model. Initially, we input the features $I \in \mathbb{R}^{C\times H\times W}$ with dimensions $3 \times 352 \times 352$. Through four encoding blocks $\mathbb{E}^i (i \in \{1, 2, 3, 4\})$, the network generates four levels of global features that encapsulate long-distance dependency information $f = \{f^1_{64\times 88\times 88}, f^2_{128\times 44\times 44}, f^3_{320\times 22\times 22}, f^4_{512\times 11\times 11}\}$, where the numerical superscript indicates the encoding block. Then, the EAEM module extracts boundary semantic information from the high-resolution and more detailed shallow features f^1 to obtain the feature f^1_E, guiding the encoder to generate a more precise saliency map with accurate boundaries. To address the issue of background interference and accurately locate the saliency map position, we sample f^1–f^3 and perform weighted attention operations in the MGWAM to generate three levels of features rich in semantic information f^1_M–f^3_M; since the deep features f^3, f^4 contain a large amount of hidden information, the DDAM is proposed to deeply mine the semantic information of deep features to obtain the characteristic f^{34}_D to adapt to the problem of missed detection of small-sized targets, while enhancing the backbone feature information of salient targets. Finally, the decoder is utilized to perform level-by-level inference on the five features to generate the final saliency detection map.

B. Edge Adaptive Enhancement Module

The shallow features generated by the encoder contain a wealth of detailed information. Based on this, the EAEM is designed to fully extract the edge detail information in feature f^1. By splitting the feature channel dimension and performing adaptive pooling operations on the features, we obtain features that contain edge details. This aids in the generation of more precise saliency map edges. We provide a comprehensive explanation of the design and implementation of EAEM, as depicted in Fig. 3.

In order to enhance the network's ability to process features and consistently emphasize edge details in each channel, we split feature f^1 along the channel dimension. This approach aids subsequent networks in better understanding and utilizing information from various channels, while also reducing redundant computations for identical features, thereby enhancing the model's overall performance. After the splitting, we obtain four features of the same dimension

$$f^i = \text{split}\left(f^1\right) \tag{1}$$

where $f^i \in \mathbb{R}^{(C/4)\times 88\times 88} (i \in \{a, b, c, d\})$, and split$(\cdot)$ denotes the splitting operation along the channel dimension.

The channel attention mechanism may fail to capture long-distance dependencies, thus resulting in the loss of spatial dimension information. Therefore, we apply a spatial attention

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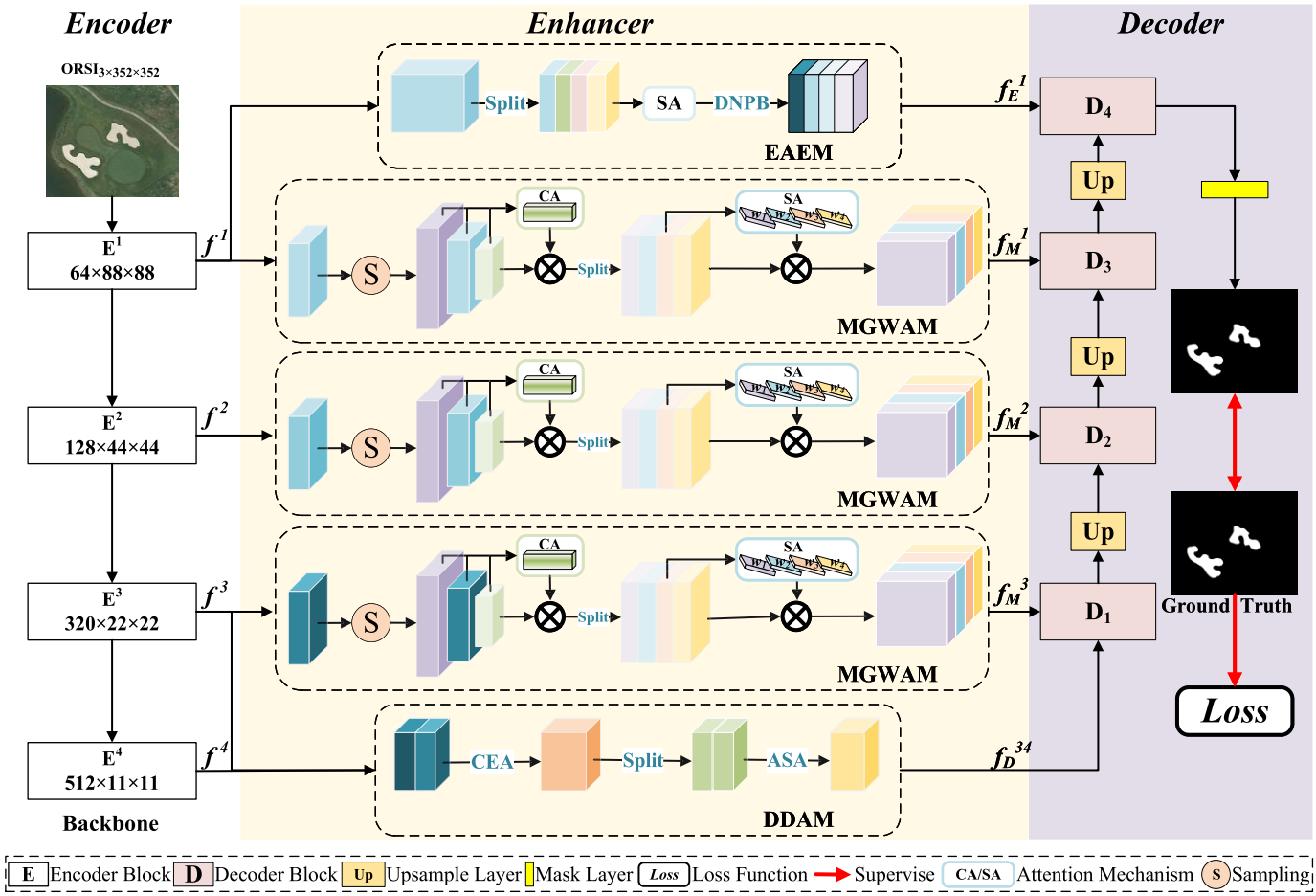


Fig. 2. Proposed overall framework of LGIPNet includes a Transformer-based feature encoder PVT-v2, three feature enhancement modules, and a saliency predictor. First, we use the Transformer-based feature extractor PVT-v2 to extract four levels of basic features. Then, we employ EAEM, MGWAM, and DDAM to process the corresponding features. Finally, the decoder generates the saliency map from the outputs of these modules.

mechanism to the aforementioned four-level features to further establish global and local connections. This method focuses on information at different spatial locations in the image, helping the model to concentrate more on the salient parts of the image. Consequently, we perform an attention operation on the features obtained after splitting. The calculations are specified as follows:

$$f_{\mathrm{sa}}^{i} = \mathrm{SA}\left(f^{i}\right) \otimes f^{i} \oplus f^{i} \tag{2}$$

where $i \in \{a, b, c, d\}$, $\mathrm{SA}(\cdot)$ is the spatial attention operation, $\otimes$ is the elementwise multiplication, and $\oplus$ is the elementwise summation.

In the EAEM module, we employ the DNPB to perform distinct pooling operations on each channel of the input features. By using a regularization function, we compute the dissimilarity between two feature maps to adaptively extract edge feature information. The architecture is depicted in the DNPB of Fig. 3. The following is a detailed description of this process:

$$F_{\mathrm{DNPB}} = \mathrm{up}_{2\times}\left(L_2\left(\mathrm{MP}\left(f^{i}\right) - \mathrm{AP}\left(f^{i}\right)\right)\right) \tag{3}$$

where $\mathrm{up}_{2\times}(\cdot)$ is the twofold upsampling operation, $L_2(\cdot)$ denotes the L_2 norm regularization, $\mathrm{AP}(\cdot)$ is the adaptive average pooling, and $\mathrm{MP}(\cdot)$ is the adaptive max pooling.

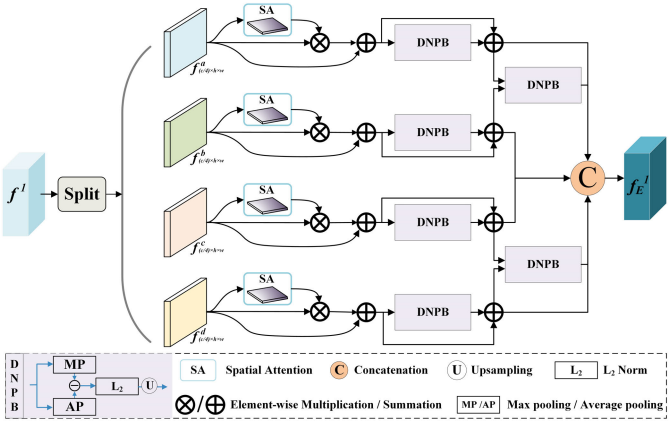


Fig. 3. We propose the detailed structure of the EAEM, which splits the features and uses attention mechanisms along with the DNPB to extract edge information from different channel features.

Max pooling (MP) retains only the maximum value in each pooling window to highlight key features, making edge information in features clearer and generating a feature map more sensitive to edge and texture information. It is highly effective in extracting high-frequency image information. Average pooling (AP) calculates the mean of all values in each window,

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suppressing local unimportant features and highlighting important ones, smoothing the feature map, reducing noise, and making it more sensitive to overall data features. In addition, differential operations on features processed by these two pooling methods yield significant edge features. Applying DNPB to feature f_{sa}^{i} results in four edge-enhanced features. The following is a detailed description of this process:

$$f_{\text{DNPB}}^{i} = F_{\text{DNPB}}\left(f_{\text{sa}}^{i}\right) \oplus f_{\text{sa}}^{i} \quad (i \in \{a, b, c, d\}). \tag{4}$$

Next, we perform differential operations between different features to explore the information between different channels. The specific process is as follows:

$$f_{\text{DNPB}}^{ab} = \text{up}\left(L_2\left(\text{MP}\left(f_{\text{DNPB}}^{a}\right) - \text{AP}\left(f_{\text{DNPB}}^{b}\right)\right)\right) \tag{5}$$

$$f_{\text{DNPB}}^{cd} = \text{up}\left(L_2\left(\text{MP}\left(f_{\text{DNPB}}^{c}\right) - \text{AP}\left(f_{\text{DNPB}}^{d}\right)\right)\right). \tag{6}$$

Finally, we concatenate all the edge-enhanced features to obtain feature f_E^1

$$f_E^1 = \text{Concat}\left(f_{\text{DNPB}}^{a}, f_{\text{DNPB}}^{b}, f_{\text{DNPB}}^{c}, f_{\text{DNPB}}^{d}, f_{\text{DNPB}}^{ab}, f_{\text{DNPB}}^{cd}\right). \tag{7}$$

C. Multiscale Grouped Weighted Attention Module

In ORSI, the persistent issues of missed detection of small targets and incomplete detection of irregular targets have been longstanding challenges. Therefore, our research focuses on the efficient extraction of multiscale information inherent in features that capture global long-distance dependencies. Extensive studies [43], [44] have underscored the vital significance of multiscale information through exploration and application, whereas other research [45], [46] has investigated the direct relationships between spatial and channel dimensions by splitting features along the channel axis.

Drawing inspiration from these findings, we have devised the MGWAM as our principal module for predicting the core of salient targets within the lower three tier features (i.e., f^1–f^3). Through sampling, we extract three features of varying sizes and perform convolutional and attention operations to explore the dimensional information across different scale feature maps, accommodating the diverse sizes of salient targets. Considering the heterogeneity of multiscale information, we further group-process features containing multiscale data along the channel axis. This strategy aims to uncover the intrinsic patterns within layer-specific features, rather than overly focusing on the interactions between layers. To ensure consistency in the final output of each feature group, we utilize a weighted attention mechanism to generate the final feature map. We provide a detailed exposition of MGWAM, with its structural depiction presented in Fig. 4.

In order to capture comprehensive multiscale information, we initially perform upsampling and downsampling on the features to acquire features of varying dimensions, denoted by $f^{\alpha} \in \mathbb{R}^{C\times(W/2)\times(H/2)}$, $f^{\beta} \in \mathbb{R}^{C\times W\times H}$, and $f^{\gamma} \in \mathbb{R}^{C\times 2W\times 2H}$

$$f^{\alpha} = \text{up}_{2\times}\left(f^{i}\right), \quad f^{\beta} = f^{i}, \quad f^{\gamma} = \text{down}_{\frac{1}{2}\times}\left(f^{i}\right) \tag{8}$$

where $\text{down}_{(1/2)\times}(\cdot)$ is the twofold downsampling operation.

Dilated convolutions are highly effective for extracting multiscale information, allowing us to expand the receptive field without reducing the image resolution. In the application of multiscale features, dilated convolutions enable the network to efficiently extract information at different scales, thereby enhancing the recognition capabilities for objects of various sizes within the image.

Therefore, by applying dilated convolutions with different dilation rates to the aforementioned three-scale features, the network can explore scale features under different receptive fields. This enables the network to capture a richer array of multiscale detail features. Through the operations described above, we obtain three tier features $f_D^i \in \mathbb{R}^{(C/4)\times W\times H}, i \in (\alpha, \gamma)$ and $f_D^{\beta} \in \mathbb{R}^{(C/4)\times W\times H}$. The specific operations are as follows:

$$f_D^{i} = \text{Dconv}\left(f^{i}\right), \quad i \in (\alpha, \beta, \gamma) \tag{9}$$

where $\text{Dconv}(\cdot)$ is a dilated convolution. Subsequently, we continue to explore the dependency information on the feature channels. Specifically, we perform channel attention operations on the aforementioned features as follows:

$$f_C^{i} = \text{CA}\left(f_D^{i}\right) \otimes f_D^{i} \oplus f_D^{i}, \quad i \in (\alpha, \beta, \gamma). \tag{10}$$

We then concatenate the multiscale features. This allows us to integrate features of different scales, forming a more comprehensive feature representation. To preserve the information of the original features while also incorporating the rich information of the multiscale features, we add the concatenated features at the element level to the initial features. The operation is as follows:

$$f^{M} = \text{Concat}\left(f_C^{\alpha}, f_C^{\beta}, f_C^{\gamma}\right) \oplus f^{i}. \tag{11}$$

Considering the differences in feature information extracted at different scales, it is, indeed, necessary to split the features for independent processing of each scale. This allows for more effective capture of information across all scales, aiding the network in a more comprehensive understanding of the image content

$$f^{j} = \text{split}\left(f^{M}\right) \quad (j \in \{a, b, c, d\}), \quad f^{j} \in \mathbb{R}^{\frac{C}{4}\times W\times H}. \tag{12}$$

Next, we perform spatial attention operations on the grouped features f^j to better understand and utilize the relationships within the feature space, resulting in the corresponding spatial attention maps. However, due to the differences in scale information in the spatial attention maps of different features after splitting, we perform weighted spatial attention operations on the split features. This ensures that while the salient target information for each channel is highlighted as much as possible, the information for each channel remains consistent. We introduce learnable weights $w_{[i]}(i \in 1, 2, 3, 4)$, and normalize the weights to ensure $\sum_{i=1}^{4} w_{[i]} = 1$ to determine the weights of different attention maps. During the training process, these weights will be adjusted to determine the importance of each scale.

Then, we perform a weighted combination of the four attention maps containing different scale information, thus obtaining a comprehensive and consistent spatial attention

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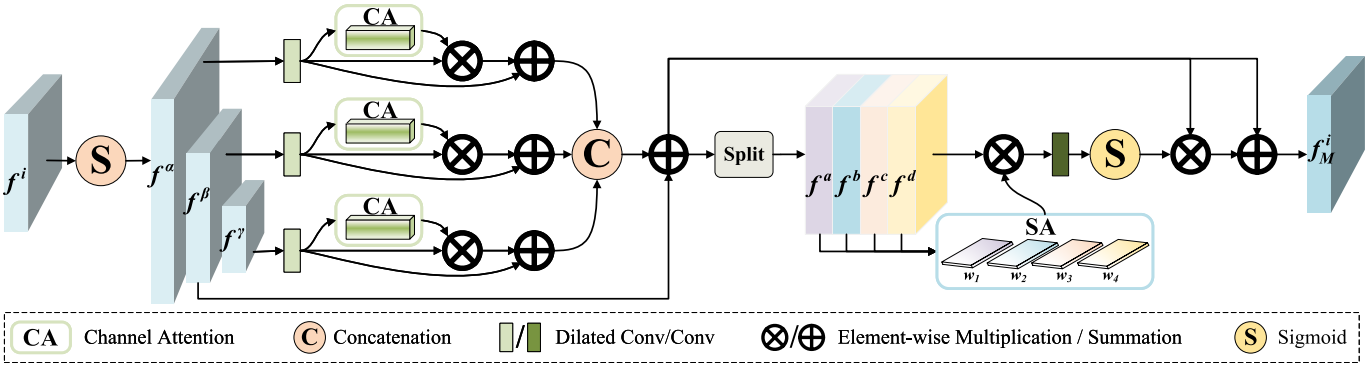


Fig. 4. We propose the detailed structure of the MGWAM, which extracts information from different scale feature channels through feature scaling and channel attention. It then aggregates spatial information through weighted spatial attention to accurately locate the position and contour of salient targets.

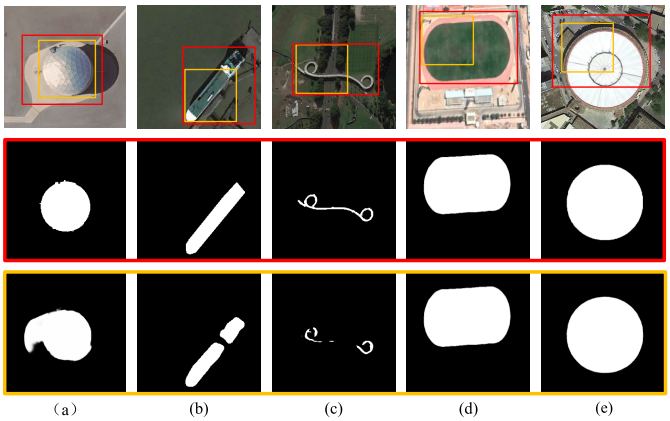


Fig. 5. Different object types require varying amounts of contextual information. The red and orang boxes represent the different receptive fields of the detectors. The second row shows the results generated by detectors with broad receptive fields, while the third row shows the results generated by detectors with limited receptive fields. (a)–(e) ORSIs with salient objects of different scales and shapes.

map. The specific operation is formulated as follows:

$$f^W = \text{conv}\left(\sum_{i=1}^{4} w_{[i]} * \text{SA}\left(f^j\right)\right). \tag{13}$$

Finally, we employ the weighted spatial attention map to spatially guide the generation of weighted features grouped on multiple scales f^W as follows:

$$f_M^i = f^M \otimes f^W \oplus f^M. \tag{14}$$

For the design process of the MGWAM, we employ a unique strategy that integrates multiscale feature information and apply an attention mechanism for weighted processing of this information, aiming to focus more on the main body of the target. Through fine control and integration of features, the MGWAM is capable of effectively distinguishing between the target and the background and extracting the most representative and critical features, thereby achieving accurate prediction of salient objects.

D. Dual-Domain Attention Module

In ORSI, salient objects, often small and complex in high-resolution aerial views, are challenging to detect due to diverse sizes and shapes. However, their surrounding context, especially shape-related, is crucial for successful detection. Salient objects of different sizes usually exist independently, with few large-size differences in one image. Background complexity varies among objects, so different approaches are needed for context information. Limited context can lead to incomplete detection of irregular objects and redundancy in complex background cases. As shown in Fig. 5, the limited receptive field in Fig. 5(a) makes target segmentation difficult, and insufficient context in Fig. 5(b) and (c) results in incomplete detection. For well-defined objects like the playground in Fig. 5(d) and the building top in Fig. 5(f), less context is needed. In short, context requirements vary by object type.

To tackle these challenges, appropriate context extraction from features is necessary. Deep features contain more global dependencies. Existing research shows that specialized modules for exploring deep feature semantics can guide result map generation. But relying on single-level features with local enhancement is insufficient. Some researchers explore cross-level feature interaction to find more relevant information, highlighting the importance of context in ORSI-SOD.

Based on the insights mentioned above, we have designed the DDAM to facilitate interaction between features at different levels. DDAM dynamically adjusts the receptive field for more effective handling of dynamic contextual information related to salient objects. Specifically, DDAM comprises two main components: CEA block and ASA block. The CEA block associates and extracts channel information from different-level features using global average and maximum dual-frequency pooling. The ASA block dynamically adjusts the feature extraction's receptive field by employing convolutional kernels of varying scales. This enables the extraction of context-specific information tailored to various types of targets. In the following text, we will delve into the detailed explanation of DDAM, with the specific structure illustrated in Fig. 6.

1) Feature Preprocessing: Considering the limited information in single-level features, we align and sample the top two-level features $f^3 \in \mathbb{R}^{320\times22\times22}$ and $f^4 \in \mathbb{R}^{512\times11\times11}$, and concatenate them along the channel dimension to obtain feature f^{34}. The specific steps are as follows:

$$f^{34} = \text{Concat}\left(f^3, \text{up}_{2\times}\left(f^4\right)\right). \tag{15}$$

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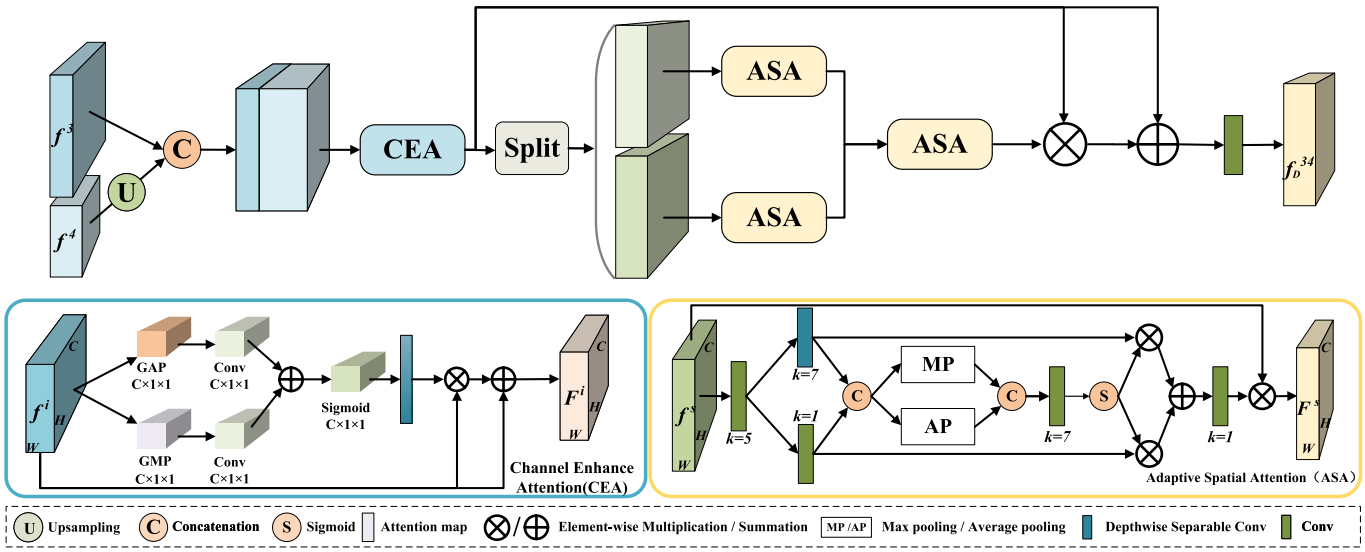


Fig. 6. We propose the detailed structure of DDAM, which uses the CEA to interact across channels without reducing dimensions, ensuring that global information is not lost while enhancing local information. The ASA extracts global information through convolutional operations with different receptive fields and captures long-distance dependency information related to the target.

2) Channel Enhancement Attention: The channel attention mechanism aids in extracting more valuable information from the backbone network and facilitates interaction between channels. However, directly applying attention mechanisms to high-level features often results in the loss of spatial information in ORSI, which hinders effective context information extraction. To address this, we design the CEA block to perform dual-frequency pooling on input features and extract and interact channelwise information. Specifically, the global AP (GAP) operation computes the average value for each channel, capturing global dependencies and enhancing local spatial dependencies. However, a single-GAP operation can sometimes blur the primary location information of salient objects. In contrast, the global maximum pooling (GMP) operation highlights the most significant features in each channel. We simultaneously apply both GAP and GMP operations to input feature $f^i \in \mathbb{R}^{C\times W\times H}$, resulting in two attention maps, X^A, $X^M \in \mathbb{R}^{C\times 1\times 1}$, which contain global spatial context information. We then perform 1-D convolutions on X^A and X^M, elementwise summation, and apply the sigmoid activation function to map the output features, yielding the attention map $X \in \mathbb{R}^{C\times 1\times 1}$ that incorporates global information

$$X^A = \mathrm{GAP}\left(f^i\right), \quad X^M = \mathrm{GMP}\left(f^i\right) \tag{16}$$

$$X = \sigma\left(\left(\mathrm{conv}_{1\times 1}\left(X^A\right) \oplus \mathrm{conv}_{1\times 1}\left(X^M\right)\right)\right) \tag{17}$$

where $\sigma(.)$ denotes the sigmoid function and $\mathrm{conv}_{1\times 1}(.)$ represents a 1×1 convolution

Next, we align the attention map X with the input features using the expand function. Finally, we combine the global attention map with the initial features by applying weights, resulting in the enhanced feature $F^i \in \mathbb{R}^{C\times H\times W}$. The specific steps are shown in the following:

$$F^i = f^i \otimes \mathrm{expand}(X) \oplus f^i. \tag{18}$$

3) Adaptive Spatial Attention: In ORSI, small objects are prone to misdetection due to insufficient use of distant contextual information, and different targets have varying demands for contextual information. Previous studies [47], [48] have highlighted the crucial role of large-receptive fields. Therefore, we designed the ASA module with different-sized convolutional kernels. By adjusting the receptive field of feature extraction, it effectively processes contextual information related to target changes, achieving ASA. Then, we extract spatial information from channel-enhanced features, realizing dual-domain feature processing in channels and space. Specifically, we decompose ordinary large-kernel convolutions into depth convolutions with gradually changing kernel sizes and dilation rates to ensure the gradual expansion of the receptive field during inference. We apply depthwise separable convolutions with kernel sizes of 5 and 7 to the input feature $f^s \in \mathbb{R}^{C\times W\times H}$ to obtain features $f^s_{k_7}$, $f^s_{k_5} \in \mathbb{R}^{C\times W\times H}$ containing contextual information from different ranges. Finally, we concatenate these features along the channel dimension to get $f^s_D \in \mathbb{R}^{2C\times W\times H}$

$$f^s_{k_5} = \mathrm{DSconv}_{k=5}\left(f^s\right), \quad f^s_{k_7} = \mathrm{DSconv}_{k=7}\left(f^s\right) \tag{19}$$

$$f^s_D = \mathrm{Concat}\left(f^s_{k_5}, f^s_{k_7}\right) \tag{20}$$

where $\mathrm{DSconv}(.)$ represents depthwise separable convolution and k denotes the scale of the convolutional kernel.

To better focus the network on spatial context regions highly relevant to salient targets, we adopt a spatial adaptive mechanism. This involves selecting spatial feature maps from convolutional kernels of different scales. We separately perform average and MP on features to adaptively extract spatial dependencies, then concatenate these pooled features. Finally, we generate the spatial attention feature $\widehat{Y} \in \mathbb{R}^{C\times W\times H}$ via large-kernel convolution. The steps are as follows:

$$\widehat{Y} = \mathrm{conv}_{k=7}\left(\mathrm{Concat}\left(\mathrm{MP}\left(f^s_D\right), \mathrm{AP}\left(f^s_D\right)\right)\right). \tag{21}$$

Next, by applying the sigmoid activation function, we obtain spatial adaptive attention maps $\widetilde{Y} \in \mathbb{R}^{2C\times W\times H}$ corresponding

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to the features extracted by each kernel. The process is formulated as follows:

$$\widetilde{Y} = \sigma\left(\widehat{Y}\right). \tag{22}$$

Afterward, the features generated by different kernels are weighted using the spatial attention maps, and the spatial attention feature $Y \in \mathbb{R}^{C\times W\times H}$ is obtained through convolution, which is calculated as shown in the following equation:

$$Y = \text{conv}_{k=1}\left(\left(f_{k_5}^s \otimes \widetilde{Y}\right) \oplus \left(f_{k_7}^s \otimes \widetilde{Y}\right)\right). \tag{23}$$

Finally, the input features are guided by feature Y to obtain the output feature $F^s \in \mathbb{R}^{C\times W\times H}$, which is calculated as shown in the following equation:

$$F^s = Y \otimes f^s. \tag{24}$$

4) Feature Enhancement Processing: In this section, we elaborate on how the DDAM processes feature using the CEA and ASA. Initially, we apply the CEA to the preprocessed feature f^{34}, focusing on channelwise processing to obtain feature $f_C^{34} \in \mathbb{R}^{832\times 22\times 22}$ as shown in the following equation:

$$f_C^{34} = \text{CEA}\left(f^{34}\right). \tag{25}$$

Afterward, we split feature f_C^{34} along the channel dimension into f_C^a, $f_C^b \in \mathbb{R}^{416\times 22\times 22}$ as shown in the following equation:

$$f_C^a, f_C^b = \text{split}\left(f_C^{34}\right). \tag{26}$$

Afterward, we utilize the ASA to thoroughly explore the spatial information present in the split features, resulting in an enhanced spatial feature f_A^a, $f_A^b \in \mathbb{R}^{416\times 22\times 22}$ as follows:

$$f_A^a = \text{ASA}\left(f_C^a\right), \quad f_A^b = \text{ASA}\left(f_C^b\right). \tag{27}$$

Next, we further explore the spatial information interdependencies between different channel features f_A^a and f_A^b using the ASA block. However, multiple convolutional operations can lead to the loss of crucial spatial information. To address this, we selectively remove the large-scale kernel convolutional operations within the ASA block and directly process features f_A^a and f_A^b. The result is a feature $f_A^{34} \in \mathbb{R}^{832\times 22\times 22}$ that has undergone dual-domain enhancement in terms of both channels and space. The specific steps are as follows:

$$f_A^{34} = \text{ASA}\left(f_A^a, f_A^b\right). \tag{28}$$

The final output of the DDAM module is achieved by aggregating the channel-enhanced features f_C^{34} and f_A^{34} through weighted fusion. This process yields the feature $f_D^{34} \in \mathbb{R}^{514\times 22\times 22}$, as depicted in the following equation:

$$f_D^{34} = f_A^{34} \otimes f_C^{34} \oplus f_C^{34}. \tag{29}$$

E. Decoder and Loss Function

1) Decoder: Inspired by the cascaded partial decoder [49], we similarly employ this decoder to guide shallow features using deep features for generating the final saliency map. For the five levels of features generated by the enhancer (f_E^1, f_M^1, f_M^2, f_M^3, and f_D^{34}), we progressively upsample and convolve the deep features to create features with expanded spatial dimensions and reduced channel numbers. Then, we sequentially cascade guide features f_E^1 and f_D^{34} that contain edge and contextual information, with features f_M^1, f_M^2 and f_M^3. Finally, through feature concatenation, downsampling, and convolutional operations, obtain the saliency map $S \in \mathbb{R}^{1\times 352\times 352}$.

2) Loss Function: Existing hybrid and complementary losses [50] have been successfully applied in SOD. In our case, we employ pixel-level binary cross-entropy (BCE) [51] loss and map-level intersection over union (IoU) [50] loss for content supervision. The overall loss function ℓ_{total} is represented as follows:

$$\ell_{\text{total}} = \ell_{\text{bce}}(S, \text{GT}) + \ell_{\text{iou}}(S, \text{GT}) \tag{30}$$

where S denotes the predicted saliency map and GT denotes the GT. The overall process of the proposed LGIPNet is shown in Algorithm 1.

Algorithm 1 LGIPNet Model for ORSI-SOD

Input: Input ORSI data $I \in \mathbb{R}^{C\times H\times W}$ and ground truth $GT \in \mathbb{R}^{C\times H\times W}$.

Output: Predicted salient maps of the test dataset.

1: Set batch size = 8, optimizer Adam (learning rate = $1e{-}4$, minimum learning rate = $1e-6$), epoches $\epsilon = 50$.
2: Generate training loader and test loader.
3: **for** $i = 1$ to ϵ **do**
4: Perform PVT-v2 module and generate initial features f^1, f^2, f^3, and f^4.
5: Input f^1 to EAEM module and generate edge enhancement features f_E^1. (See Equation (7))
6: Input f^1, f^2, and f^3 to MGWAM module and generate information enhancement features f_M^1-f_M^3.(See Equation (14))
7: Input f^3, f^4 to DDAM module and generate semantic enhancement features f_D^{34}. (See Equation (29))
8: Sequentially fuse the enhanced features to generate the final saliency map through supervisory guidance.
9: **end for**
10: Apply the trained model to the test dataset for predicted saliency maps.

IV. Experiments

A. Experimental Protocol

1) Evaluation Metrics: The performance of our method and the SOTA methods is assessed quantitatively using five widely used metrics [63], including the structural metric (S_α) [64], the (maximum, average, and adaptive) F-measure (F_β, $\beta^2 = 0.3$) [65], the (maximum, average and adaptive) augmented localization metric (E_ξ) [66], the mean absolute error (MAE) [67], the precision–recall curve (PR) [68], and the F-measure curves.

2) Datasets: We have trained and validated our network on the following three mainstream datasets for ORSI-SOD. The ORSSD [16] dataset contains 800 challenging ORSI and their corresponding GT images, of which 600 are used for training and 200 for testing. The EORSSD [17] dataset encompasses all the data from the ORSSD dataset and builds

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TABLE I

QUANTITATIVE COMPARISON OF OUR METHOD WITH 22 OTHER STATE-OF-THE-ART METHODS, INCLUDING THREE TRADITIONAL ORSI-SOD METHODS, 15 CNN-BASED ORSI-SOD METHODS, AND FOUR TRANSFORMER-BASED ORSI-SOD METHODS, ON TWO PUBLICLY POPULAR DATASETS WITH S_MEASURE, MAXIMUM, MEAN AND ADAPTIVE F-MEASURE, MAXIMUM, MEAN AND ADAPTIVE E_MEASURE, AND MAE. ↑ AND ↓ DENOTE LARGER AND SMALLER, RESPECTIVELY. THE TOP THREE RESULTS ARE MARKED IN RED, BLUE, AND GREEN, RESPECTIVELY

Type	Methods	SPeed ($FPS\uparrow$)	Params ($M\downarrow$)	FLOPs ($G\downarrow$)	EORSSD [17]								ORSSD [16]							
					$S_\alpha\uparrow$	$F_\beta^{max}\uparrow$	$F_\beta^{mean}\uparrow$	$F_\beta^{adp}\uparrow$	$E_\xi^{max}\uparrow$	$E_\xi^{mean}\uparrow$	$E_\xi^{adp}\uparrow$	$MAE\downarrow$	$S_\alpha\uparrow$	$F_\beta^{max}\uparrow$	$F_\beta^{mean}\uparrow$	$F_\beta^{adp}\uparrow$	$E_\xi^{max}\uparrow$	$E_\xi^{mean}\uparrow$	$E_\xi^{adp}\uparrow$	$Mea\downarrow$
T.R.	VOS_{18} [52]	-	-	-	0.5083	0.2776	0.2115	0.1843	0.5999	0.4895	0.4776	0.2095	0.5366	0.3417	0.2717	0.2633	0.6514	0.5352	0.5826	0.2151
	CMC_{19} [53]	-	-	-	0.5800	0.3272	0.2696	0.2009	0.6818	0.5900	0.4890	0.1057	0.6033	0.3913	0.3454	0.3108	0.7064	0.6417	0.5996	0.1267
	$SMFF_{19}$ [54]	-	-	-	0.5405	0.5208	0.3011	0.2089	0.7768	0.5210	0.5019	0.1434	0.5312	0.4417	0.2684	0.2496	0.7402	0.492	0.5676	0.1854
C.R.	$LVNet_{19}$ [16]	-	-	-	0.8630	0.7794	0.7328	0.6284	0.9254	0.8801	0.8445	0.0146	0.8815	0.8263	0.7995	0.7506	0.9456	0.9259	0.9195	0.0207
	$DAFNet_{21}$ [17]	26	29.4	68.5	0.9166	0.8614	0.7845	0.6427	0.9861	0.9291	0.8446	0.0060	0.9191	0.8928	0.8511	0.7876	0.9771	0.9539	0.9360	0.0113
	$RRNet_{21}$ [55]	16	67.6	255.5	0.9266	0.8785	0.8394	0.7416	0.9684	0.9442	0.9150	0.0074	0.9328	0.8964	0.8720	0.8249	0.9679	0.9547	0.9498	0.0112
	$EMFINet_{22}$ [20]	25	107.3	480.9	0.9290	0.8720	0.8486	0.7984	0.9711	0.9604	0.9501	0.0084	0.9366	0.9002	0.8856	0.8617	0.9737	0.9671	0.9663	0.0109
	$ERPNet_{22}$ [27]	50	56.5	87.2	0.9210	0.8632	0.8304	0.7554	0.9603	0.9401	0.9228	0.0089	0.9254	0.8974	0.8745	0.8356	0.9710	0.9566	0.9520	0.0135
	$MJRBM_{22}$ [21]	32	43.5	95.7	0.9197	0.8656	0.8239	0.7066	0.9646	0.9350	0.8897	0.0099	0.9204	0.8842	0.8566	0.8022	0.9623	0.9415	0.9328	0.0163
	$MCCNet_{22}$ [18]	95	67.7	112.8	0.9327	0.8904	0.8604	0.8137	0.9755	0.9685	0.9538	0.0066	0.9437	0.9155	0.9054	0.8957	0.9800	0.9758	0.9735	0.0087
	$ACCONet_{23}$ [28]	81	102.6	180.0	0.9290	0.8837	0.8552	0.7969	0.9727	0.9653	0.9450	0.0074	0.9437	0.9149	0.8971	0.8806	0.9796	0.9754	0.9721	0.0088
	$BAFSNet_{23}$ [56]	45	31.0	16.0	0.9287	0.8708	0.8583	0.8477	0.9772	0.9734	0.9441	0.0060	0.9385	0.9136	0.9017	0.8982	0.9819	0.9785	0.9615	0.0079
	$SRAL_{23}$ [57]	13	31.9	234.0	0.9250	0.8712	0.8488	0.8042	0.9703	0.9614	0.9215	0.0069	0.9322	0.9007	0.8783	0.8534	0.9777	0.9685	0.9463	0.0107
	$CRNet_{23}$ [29]	17	67.6	255.0	0.9370	0.8873	0.8653	0.8140	0.9743	0.9618	0.9563	0.0063	0.9389	0.9107	0.8951	0.8695	0.9793	0.9687	0.9711	0.0091
	$AESINet_{23}$ [58]	25	41.1	55.0	0.9347	0.8792	0.8496	0.7909	0.9757	0.9646	0.9297	0.0064	0.9455	0.9160	0.8962	0.8640	0.9814	0.9741	0.9538	0.0085
	$SFANet_{24}$ [59]	21	25.1	7.7	0.9349	0.8833	0.8680	0.8492	0.9769	0.9726	0.9681	0.0058	0.9453	0.9192	0.9063	0.8984	0.9830	0.9789	0.9800	0.0077
	$MCPNet_{24}$ [60]	39	25.1	86.6	0.9373	0.8868	0.8620	0.8077	0.9765	0.9657	0.9527	0.0070	0.9433	0.9135	0.8979	0.8726	0.9807	0.9736	0.9724	0.0090
	$WeightNet_{24}$ [61]	-	99.1	221.5	0.9384	0.8907	0.8764	0.8472	0.9756	0.9710	0.9659	0.0063	0.9481	0.9188	0.9072	0.8971	0.9817	0.9779	0.9787	0.0075
	LGIPNet-Res	23	65.8	41.4	0.9347	0.8859	0.8729	0.8604	0.9760	0.9705	0.9707	0.0061	0.9494	0.9232	0.9060	0.8933	0.9862	0.9808	0.9794	0.0076
	LGIPNet-VGG	27	33.9	63.8	0.9340	0.8836	0.8700	0.8560	0.9761	0.9710	0.9700	0.0065	0.9453	0.9149	0.9027	0.8924	0.9806	0.9758	0.9755	0.0082
T-C.R.	$HFANet_{22}$ [19]	27	60.5	68.3	0.9380	0.8876	0.8681	0.8365	0.9740	0.9679	0.9644	0.0070	0.9399	0.9112	0.8981	0.8819	0.9770	0.9712	0.9722	0.0092
	$ASTT_{23}$ [62]	13	43.1	23.4	0.9253	0.8741	0.8297	0.7534	0.9757	0.9584	0.9247	0.0059	0.9348	0.9061	0.8811	0.8456	0.9795	0.9699	0.9587	0.0094
	$TLCKD\text{-}Net_{24}$ [40]	38	50.0	32.0	0.9352	0.8843	0.8535	0.7969	0.9788	0.9660	0.9214	0.0056	0.9421	0.9114	0.8947	0.8719	0.9794	0.9709	0.9497	0.0082
	$ADSTNet_{24}$ [41]	40	62.1	27.7	0.9311	0.8804	0.8716	0.8532	0.9769	0.9709	0.9681	0.0065	0.9379	0.9124	0.9042	0.8979	0.9807	0.9740	0.9785	0.0086
	LGIPNet-PVT	36	52.1	47.0	0.9392	0.8924	0.8770	0.8623	0.9806	0.9753	0.9719	0.0053	0.9507	0.9266	0.9129	0.9027	0.9876	0.9829	0.9809	0.0066

T.R.: Traditional ORSI-SOD method; C.R.: CNN-based ORSI-SOD method; T-C.R.: ORSI-SOD method based on fusion of CNN and Transformer;Abbreviated subscripts of published journals correspond to the year of publication.

upon it. It has a total of 2000 ORSI, of which 1400 are used for training and 600 for testing. The ORSI-4199 [21] dataset is a highly challenging dataset with diverse targets, varying sizes, and complex backgrounds. The dataset includes a total of 4199 ORSI, of which 2000 are used for training and 2199 for testing.

3) Implementation Details: All experiments are implemented based on PyTorch, using NVIDIA GeForce RTX 2080TI GPU (with 11-GB memory) for model training computation. The image size for both training and testing is 352 × 352. The PVT model is used as our initial encoder and loads the initial weights, and other convolutional layers are initialized using the Kaiming-normal strategy. We use rotation and flipping to augment the dataset, preventing model overfitting during training. Our model uses the Adam optimizer [69] for 50 epochs of training, with a batch size of 8, an initial learning rate of $1e^{-4}$, and a weight decay rate of 0.01.

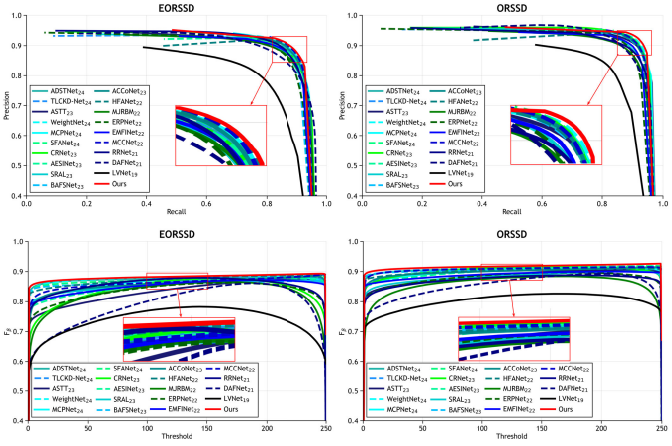


Fig. 7. Quantitative comparison of PR curves (row 1) and F-measure curves (row 2) on the EORSSD and ORSSD datasets. Our method is highlighted in red. The closer the PR curve is to the upper right corner, the better the performance of the method. The more areas where the F-measure curve is higher than the curves of other methods, the better the performance of the method.

B. Comparison With SOTA

We conducted a comprehensive evaluation and comparison of our proposed LGIPNet with 22 mainstream ORSI-SOD methods on two well-known ORSI-SOD datasets, EORSSD and ORSSD. We selected the 16 most advanced methods for comparative evaluation with our method on the challenging ORSI-4199 dataset. Specifically, this includes three

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TABLE II

COMPARISON OF SOTA ON THE ORSI-4199 DATASET, DATA COMPARISON AMONG 16 EVALUATION METHODS. THE TOP THREE RESULTS ARE MARKED IN RED, BLUE, AND GREEN, RESPECTIVELY

Type	Methods	ORSI-4199							
		$S_\alpha \uparrow$	$F_\beta^{max} \uparrow$	$F_\beta^{mean} \uparrow$	$F_\beta^{adp} \uparrow$	$E_\xi^{max} \uparrow$	$E_\xi^{mean} \uparrow$	$E_\xi^{adp} \uparrow$	$MAE \downarrow$
C.R.	DAFNet$_{21}$	0.8460	0.8332	0.7987	0.7374	0.9203	0.8813	0.8605	0.0443
	RRNet$_{21}$	0.7399	0.7314	0.7088	0.7660	0.8039	0.7670	0.8277	0.0945
	EMFINet$_{22}$	0.8675	0.8584	0.8479	0.8186	0.9340	0.9257	0.9135	0.0330
	ERPNet$_{22}$	0.8670	0.8553	0.8374	0.8024	0.9290	0.9149	0.9039	0.0357
	MJRBM$_{22}$	0.8593	0.8493	0.8309	0.7995	0.9311	0.9102	0.9088	0.0374
	MCCNet$_{22}$	0.8746	0.8690	0.8630	0.8592	0.9413	0.9348	0.9401	0.0316
	ACCONet$_{23}$	0.8775	0.8686	0.8620	0.8581	0.9412	0.9342	0.9402	0.0314
	SRAL$_{23}$	0.8743	0.8684	0.8588	0.8398	0.9445	0.9358	0.9318	0.0305
	CRNet$_{23}$	0.8699	0.8689	0.8570	0.8292	0.9397	0.9244	0.9188	0.0337
	AESINet$_{23}$	0.8755	0.8726	0.8625	0.8243	0.9459	0.9372	0.9194	0.0305
	SFANet$_{24}$	0.8761	0.8710	0.8659	0.8647	0.9447	0.9385	0.9418	0.0292
	MCPNet$_{24}$	0.8736	0.8667	0.8563	0.8458	0.9402	0.9331	0.9359	0.0324
	WeightNet$_{24}$	0.8786	0.8695	0.8635	0.8613	0.9420	0.9333	0.9380	0.0297
	LGIPNet-Res	0.8833	0.8769	0.8729	0.8361	0.9483	0.9414	0.9227	0.0283
	LGIPNet-VGG	0.8847	0.8782	0.8673	0.8280	0.9476	0.9424	0.9216	0.0288
T-C.R.	HFANet$_{22}$	0.8767	0.8700	0.8624	0.8323	0.9431	0.9336	0.9191	0.0314
	ASTT$_{23}$	0.8828	0.8788	0.8666	0.8527	0.9512	0.9428	0.9388	0.0283
	ADSTNet$_{24}$	0.8710	0.8698	0.8653	0.8655	0.9433	0.9356	0.9413	0.0318
	LGIPNet-PVT	0.8854	0.8791	0.8668	0.8268	0.9499	0.9432	0.9213	0.0276

traditional ORSI-SOD methods (VOS [52], CMC [53], and SMFF [54]), 15 CNN-based ORSI-SOD methods (LVNet [16], DAFNet [17], RRNet [55], EMFINet [20], EPRNet [27], MJRBM [21], MCCNet [18], ACCoNet [28], BAFSNet [56], SRAL [57], CRNet [29], AESINet [58], SFANet [59], MCP-Net [60], and WeightNet [61]), and four methods based on the combination of CNN and Transformer (HFANet [19], ASTT [62], TLCKDNet [40], and ADSTNet [41]), to verify the effectiveness of our network. Here, for a comprehensive comparison, in addition to the LGIPNet with the PVT backbone, we also provide the LGIPNet based on VGG [70] and ResNet [71], which are named LGIPNet-Res and LGIPNet-VGG, respectively. The saliency maps of the comparison methods were all obtained by retraining with open-source code and the same training set and parameters as in this study. For methods without open-source code, we compared them using the saliency maps provided by the methods.

1) Quantitative Comparison on the EORSSD and ORSSD Datasets: In Table I, we report the quantitative comparison results of our method and the other 22 SOTA methods on EORSSD and ORSSD. As can be seen from the highlighted metrics in Table I, our method has a clear advantage on the EORSSD and ORSSD datasets. Specifically, on the EORSSD dataset, our LGIPNet outperforms other methods in seven out of eight metrics, with $E_\xi^{\max}$ being slightly lower than the comparison method and ranking second. Among these, our method excels in three different dimensions of the F-measure ($F_\beta^{\max}$, F_β^{mean}, F_β^{adp}) and two different dimensions of the E-measure (E_ξ^{mean}, $E_\xi^{\max}$), outperforming other comparison methods and significantly outpacing the second-best method.

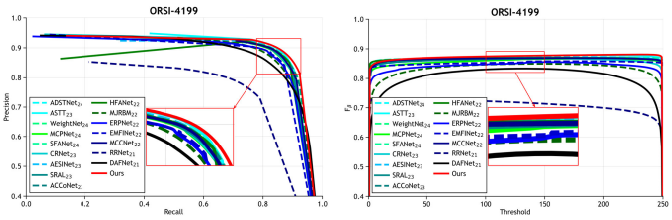


Fig. 8. Quantitative comparison of PR curves and F-measure curves on the ORSI-4199 datasets. Our method is highlighted in red.

TABLE III

COMPARISON OF RESULTS ON THE CHALLENGING SCENES SSO, CS, AND NSO IN THE CHALLENGING ORSI-4199 DATASET. THE BEST RESULTS ARE INDICATED IN BOLD

Methods	SSO				CS				NSO			
	$S_\alpha \uparrow$	$F_\beta^{max} \uparrow$	$E_\xi^{max} \uparrow$	$MAE \downarrow$	$S_\alpha \uparrow$	$F_\beta^{max} \uparrow$	$E_\xi^{max} \uparrow$	$MAE \downarrow$	$S_\alpha \uparrow$	$F_\beta^{max} \uparrow$	$E_\xi^{max} \uparrow$	$MAE \downarrow$
DAFNet$_{21}$	0.8326	0.8256	0.8936	0.0765	0.8365	0.8264	0.9035	0.0495	0.8634	0.8589	0.8900	0.0611
MCCNet$_{22}$	0.8752	0.8623	0.9218	0.0543	0.9059	0.8966	0.9256	0.0463	0.8732	0.8681	0.8935	0.0520
HFANet$_{22}$	0.7955	0.7868	0.8279	0.1156	0.8277	0.8144	0.8868	0.0534	08856	0.8802	0.9066	0.0506
AESINet$_{23}$	0.8656	0.8453	0.9167	0.0567	0.9126	0.9020	0.9313	0.0412	0.9011	0.8896	0.9213	0.0456
ATSS$_{23}$	0.8576	0.8514	0.9135	0.0603	0.9162	0.9102	0.9214	0.0382	0.9166	**0.9127**	0.9368	0.0308
SFANet$_{24}$	0.8734	0.8639	0.9246	0.0547	0.9037	0.8869	0.9116	0.0477	0.9007	0.8946	0.9201	0.0483
ADSTNet$_{24}$	0.8637	0.8598	0.9383	0.0588	0.9155	0.9064	**0.9523**	0.0396	0.8926	0.8849	0.9146	0.0492
Ours	**0.8823**	**0.8662**	**0.9553**	**0.0432**	**0.9233**	**0.9147**	0.9521	**0.0368**	**0.9201**	0.9102	**0.9424**	**0.0298**

This is obvious to show that our method is not only effective for large-scale obvious targets but also very effective for difficult-to-detect small targets. In S_α and MAE, LGIPNet leads the second best method by 0.08% (0.9392 versus 0.9384) and 0.03% (0.0053 versus 0.0056) respectively. Similarly, it performs excellently on the ORSSD, a small-sample dataset. On this dataset, our LGIPNet achieves the best results in all metrics. Notably, S_α and MAE are significantly ahead of the second-best method, with MAE (0.0066 versus 0.0075) being the only metric to achieve a score below 0.007, marking a substantial lead.

Since our modules are specifically designed for the global characteristics of the Transformer, although the performance of our two CNN-based variants, LGIPNet-VGG and LGIPNet-Res, is not as good as that of the LGIPNet based on PVT, they still outperform other comparison methods in multiple metrics on the two datasets. They rank first and second in multiple metrics and can be comparable to state-of-the-art methods, such as MCPNet, WeightNet, and ADSTNet. To further illustrate the effectiveness of our method, we plot the PR curve and F-measure curve of 19 SOTA methods other than three traditional methods in Fig. 7 on the EORSSD and ORSSD datasets. We can find that the PR curve of our model is closest to the coordinate (1, 1), and the area below the F-measure curve of our model is also the largest. In a word, our method surpasses other SOD models under the PR curve and F-measure curve indicators.

2) Quantitative Comparison on the ORSI-4199 Dataset: In Table II, we report the quantitative comparison results of our method alongside 16 other SOTA methods on the ORSI-4199 dataset. As indicated in the table, our method demonstrates a clear superiority on the challenging ORSI-4199 dataset, with four evaluation metrics outperforming the competing methods. Notably, S_α and MAE are significantly

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ahead of the second-best method, with MAE (0.0276 versus 0.0283) being the only metric to achieve a score below 0.028, marking a substantial lead. The performance of LGIPNet-VGG and LGIPNet-Res on the ORSI-4199 dataset is also excellent. However, there is still a gap compared with LGIP based on PVT, which further confirms that our modules are specifically designed for the global characteristics of the Transformer.

To further verify the contributions of our work, we conducted experimental analyses on different challenging datasets within ORSI-4199, including the small-salient object (SSO) dataset, the complex scene object (CS) dataset, and the narrow and slender salient object (NSO) dataset. As shown in Table III, we compared our method with seven mainstream methods that target the above challenges on these three challenging datasets. Our method outperforms the comparative methods in all indicators on the SSO dataset, achieving phased advantages in each indicator. We also lead other comparative methods in three indicators on both CS and NSO datasets.

This demonstrates that our method yields better detection outcomes when faced with a variety of challenging detection targets. Similarly, we have illustrated the PR curve and F-measure curve of the aforementioned comparison methods in Fig. 8 using the ORSI-4199 dataset. Following quantitative analysis across the three distinct datasets, it is evident that our method exhibits performance advantages on all of these datasets and significantly outperforms methods based on full CNN and traditional methods, thereby confirming the value of each module designed by our Transformer-based backbone.

3) Qualitative Comparison: In Fig. 9, we present a visual comparison of our method with other deep learning-based approaches. We perform this visual comparison across various challenge types within the ORSI dataset. In Fig. 9, the first column displays several representative ORSI images, while the second and third columns show the corresponding GT and the prediction results produced by our method, respectively. From this, we can directly observe the alignment between our prediction results and the GT. Below, we discuss the advantages of our method across four distinct dimensions.

a) Multismall target ORSI: For the detection of multiple small targets, our method demonstrates superior capability in matching the GT images, both in terms of the number of targets and the precision of their specific contours, when compared to other methods. Other approaches frequently encounter issues where small targets are either missed or falsely detected. While some methods may achieve accuracy in the count of targets, they often struggle to precisely delineate the contours of small targets, leading to instances of redundant detection. In contrast, our method is more adept at tackling the detection of multiple small targets within complex backgrounds. This is primarily attributed to our network's ability to effectively mine deep semantic information and accurately detect the boundaries of salient targets in ORSI images. It accurately locates all tiny protruding objects, exhibiting robust scale adaptability.

b) Complex background target ORSI: In the figure, we have visualized a comparison for ORSI that are large in size and have complex backgrounds. Other methods cannot fully detect the targets in large-scale and complex-background ORSI, leading to the omission of some target content in the image and the detection of irrelevant parts. This may be due to incomplete extraction of target information and excessive extraction of background information. In contrast, our method can not only effectively suppress the noise in the complex background but also ensure the detection of the overall contour of large-sized targets. This is due to our method's ability to extract long-distance dependent information in ORSI and suppress background noise.

c) Irregular target ORSI: We further analyze the ORSI for irregular targets. The detection targets have complex shapes, and we can clearly see the advantages of our results compared to those of other methods. For example, in the seventh row, where the task is to detect multiple cars side-by-side, only our method accurately identifies all the vehicles. In contrast, there is a significant discrepancy between the results of other methods and the GT. Our method successfully detects irregular targets from complex backgrounds along their edges, whereas the comparison methods exhibit omissions in target detection and have blurred edge detections.

d) Edge complex target ORSI: At the bottom of the figure, we analyze the ORSI of targets with complex edges. This type of image effectively highlights the strengths of our method. These images mostly have targets of varying sizes and shapes, and it is challenging to distinguish the target edges from the background, posing a highly complex task. As visualized in the figure, it is evident that other methods struggle to overcome the interference of background noise at the target edges. There are instances where the edge background is misidentified, leading to an excessive number of redundant detections in the prediction image. For example, in the visualization comparison of the last row, other methods fail to accurately locate the target, let alone segment the correct target edge with precision. In contrast, LGIPNet is the sole method that can accurately and completely detect the target without any superfluous detections. This is due to the design of our method's three distinct modules for edge enhancement, background suppression, and context information extraction.

C. Ablation Study

In this section, we conduct ablation experiments on the EORSSD dataset to verify the effectiveness of each module in LGIPNet and each component in the three modules. Therefore, we have experimentally verified the importance of the three proposed modules in optimizing SOD. The specific work is as follows: 1) the individual contribution of the three modules to the network; 2) the effectiveness of each component in EAEM; 3) the effectiveness of each component in MGWAM; and 4) the effectiveness of each component in DDAM. For each experiment, we retrain the model with the same parameter settings and training data.

1) Contribution of Each Module: To investigate the individual contributions of the proposed modules, namely, EAEM, MGWAM, and DDAM, we designed a variety of combinations of these three modules, resulting in seven variants: 1) Baseline; 2) Baseline + EAEM; 3) Baseline + MGWAM; 4) Baseline + DDAM; 5) Baseline + MGWAM + DDAM; 6) Baseline +

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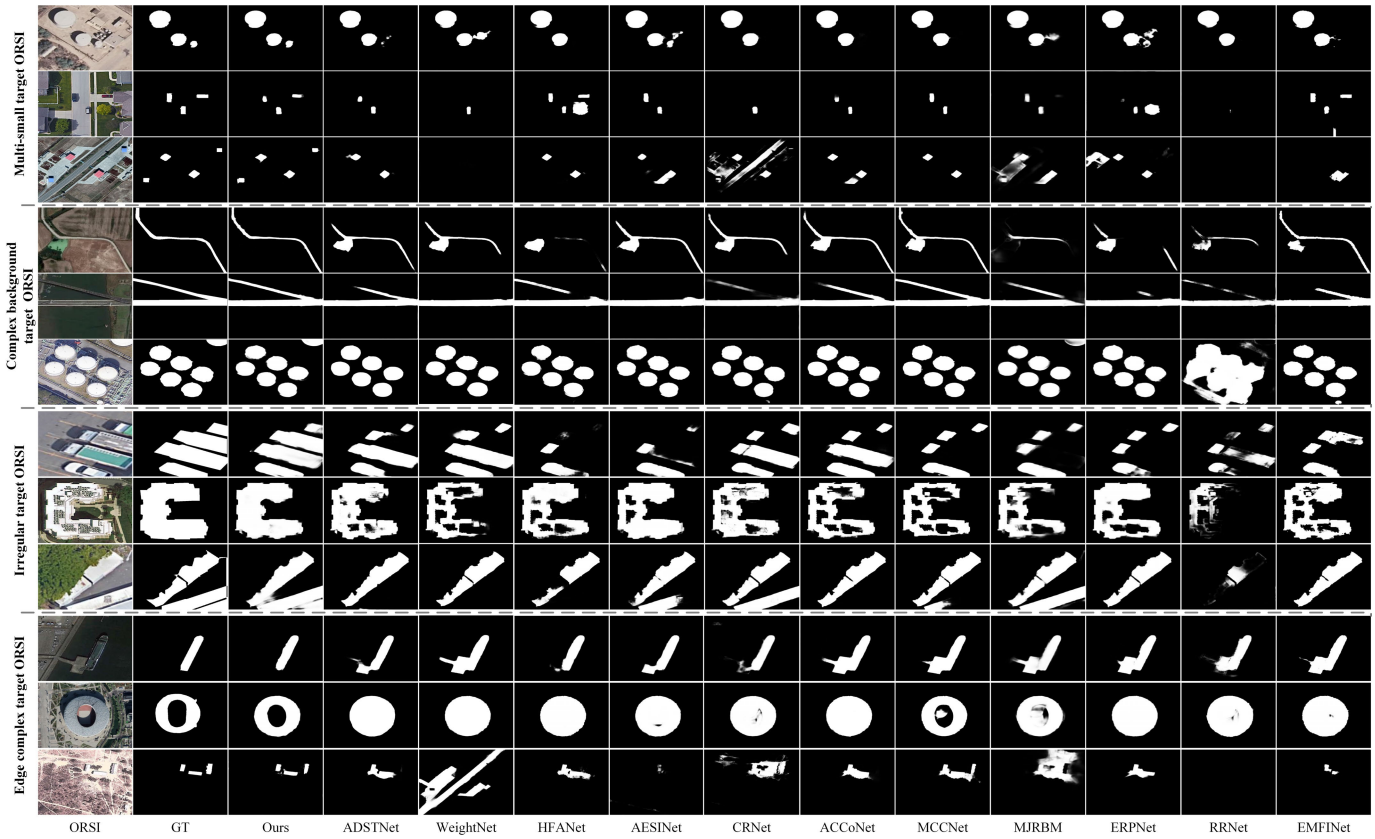


Fig. 9. Our method (LGIPNet) is visually compared with 11 SOTA methods across four categories of data with different challenges.

TABLE IV

ABLATION RESULTS OF EVALUATING THE INDIVIDUAL CONTRIBUTION OF EACH MODULE IN LGIPNET. THE BEST ONE IN EACH COLUMN IS BOLD

No.	Modules				EORSSD								
	Baseline	EAEM	MGWAM	DDAM	$S_\alpha \uparrow$	$F_\beta^{max} \uparrow$	$F_\beta^{mean} \uparrow$	$F_\beta^{adp} \uparrow$	$E_\xi^{max} \uparrow$	$E_\xi^{mean} \uparrow$	$E_\xi^{adp} \uparrow$	$MAE \downarrow$
1	✓				0.9258	0.8729	0.8592	0.8355	0.9707	0.9444	0.9412	0.0079
2	✓	✓			0.9352	0.8838	0.8632	0.8469	0.9770	0.9525	0.9579	0.0066
3	✓		✓		0.9349	0.8856	0.8688	0.8576	0.9769	0.9633	0.9634	0.0062
4	✓			✓	0.9351	0.8858	0.8653	0.8559	0.9774	0.9584	0.9625	0.0070
5	✓		✓	✓	0.9388	0.8912	0.8722	0.8593	0.9796	0.9720	0.9702	0.0059
6	✓	✓		✓	0.9375	0.8868	0.8692	0.8586	0.9790	0.9691	0.9687	0.0058
7	✓	✓	✓		0.9386	0.8916	0.8726	0.8603	0.9801	0.9711	0.9695	0.0056
8	✓	✓	✓	✓	**0.9392**	**0.8924**	**0.8770**	**0.8623**	**0.9806**	**0.9753**	**0.9719**	**0.0053**

EAEM + DDAM; and 7) Baseline + EAEM + MGWAM. The quantitative results are reported in Table IV.

a) Baseline: We use the PVT V2 prediction network as the experimental baseline, removing all proposed modules. We perform element-wise summation on the features generated by the backbone for fusion, and use the same decoder in the model architecture diagram to generate the final result. We also apply the same loss function to the output of the decoder to supervise the output of the final result. As shown in No. 1 of Table IV, on EORSSD, the baseline performance of $F_\beta^{\max}$ is 0.8729, F_β^{mean} is 0.8592, F_β^{adp} is 0.8355, $E_\xi^{\max}$ is 0.9707, E_ξ^{mean} is 0.9444, E_ξ^{adp} is 0.9412, MAE is 0.0079, and S_α is 0.9258.

b) Effectiveness of EAEM: We set up different experiments to verify the effectiveness of EAEM. First, based on the baseline network, we only apply EAEM to the lowest-level features f^1 generated by the encoder to verify the effectiveness of this module for edge processing. We fuse the features processed by EAEM with other high-level features generated by the encoder at the element level and use the same decoder to generate the final result. As shown in No. 2 of Table IV, through the optimization of the resulting edge by EAEM, the experimental result is significantly better than the baseline. Compared with it, the performance of $F_\beta^{\max}$ has increased by 1.09%, $E_\xi^{\max}$ has increased by 0.63%, MAE's performance has increased by 0.13%, and S_α has increased by 0.94%, all other indicators have also been significantly improved. Second,

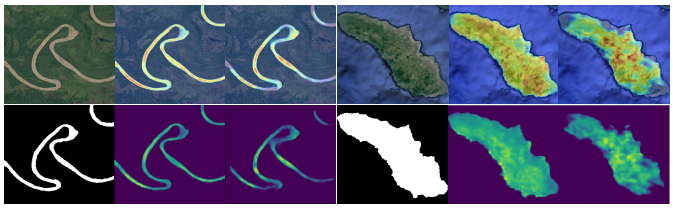


Fig. 10. Visualization results of EAEM ablation experiment. The first row shows the attention feature map visualization of different models. The second row shows the feature masks of different models.

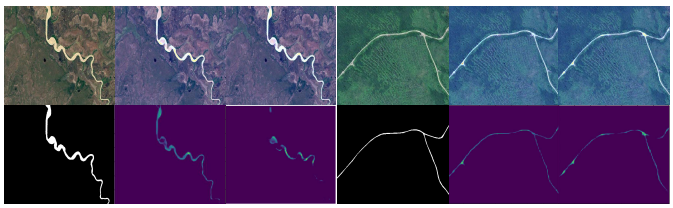


Fig. 11. Visualization results of MGWAM ablation experiment. The first row shows the attention feature map visualization of different models. The second row shows the feature masks of different models.

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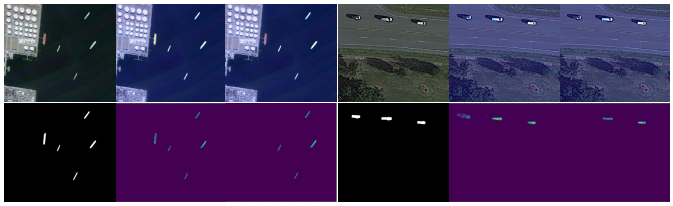


Fig. 12. Visualization results of DDAM ablation experiment. The first row shows the attention feature map visualization of different models. The second row shows the feature masks of different models.

we set up an experiment that only removes EAEM (No. 5). We apply the other two modules to this network, do not process the lowest-level features, directly fuse the results generated by the rest of the modules, and generate the final result through the decoder. As shown in No. 5 of Table IV, it can be clearly seen that EAEM enhances our method. Focusing on the comparison of the representative indicators, the performance of $F_\beta^{\max}$ has increased by 0.11%, $E_\xi^{\max}$ has increased by 0.10%, MAE's performance has increased by 0.06%, and S-measure has increased by 0.04%. In addition, we visually compare the ablation experiments of EAEM. As shown in Fig. 10, the first and second rows of the figure, respectively, show the attention maps and feature mask results of different experiments. It can be clearly seen that the experimental results of the fusion EAEM can focus more attention on the edge, and the generated feature masks also have more distinct edges. All of the above demonstrates that EAEM is effective.

c) Effectiveness of MGWAM: We set up different experiments to verify the effectiveness of MGWAM. First, based on the baseline network, we apply MGWAM to the first three levels of features f^1–f^3 generated by the encoder to try to verify the effectiveness of this module. Then, we fuse the features generated by MGWAM with the highest level features generated by the encoder at the element level and use the same decoder to generate the final result. As shown in No. 3 of Table IV, the experimental results are significantly better than the baseline. Compared with it, the performance of various indicators of the F-measure has increased by approximately 0.96% to 2.21%, the performance of various indicators of the E-measure has increased by around 0.62%–2.22%, MAE's performance has increased by 0.17%, and S_α has increased by 0.91%. Second, we set up an experiment that only removes MGWAM (No. 6). We apply the other two modules to this network, do not process the middle two levels of features, directly fuse the results generated by the rest of the modules, and generate the final result through the decoder. As shown in No. 6 of Table IV, the effectiveness of MGWAM can be found. MGWAM has improved the model's performance in $F_\beta^{\max}$, $E_\xi^{\max}$, MAE, and S_α by 0.55%, 0.16%, 0.05%, and 0.07% respectively, all other indicators have also been significantly improved. We also visually compare the ablation experiments of MGWAM. As shown in Fig. 11, according to the comparison of attention maps and feature mask results, it can be clearly seen that the experimental results of the fusion MGWAM can focus more attention on the overall target, and the generated feature masks are relatively more complete. All of the above demonstrates that MGWAM is effective.

TABLE V

ABLATION RESULTS OF EVALUATING THE EFFECTIVENESS OF EACH COMPONENT IN THE MODULE. THE BEST ONE IN EACH COLUMN IS BOLD

Modules		EORSSD							
		$S_\alpha \uparrow$	$F_\beta^{max} \uparrow$	$F_\beta^{mean} \uparrow$	$F_\beta^{adp} \uparrow$	$E_\xi^{max} \uparrow$	$E_\xi^{mean} \uparrow$	$E_\xi^{adp} \uparrow$	$MAE \downarrow$
Baseline+EAEM		**0.9352**	**0.8838**	**0.8632**	**0.8469**	**0.9770**	**0.9525**	**0.9579**	**0.0066**
EAEM	*w/o* split	0.9349	0.8822	0.8618	0.8451	0.9765	0.9506	0.9559	0.0067
	w/o splitSA	0.9320	0.8818	0.8616	0.8452	0.9757	0.9473	0.9528	0.0069
	w/o DNPB	0.9312	0.8768	0.8612	0.8436	0.9749	0.9486	0.9506	0.0071
Baseline+MGWAM		**0.9349**	**0.8856**	**0.8688**	**0.8576**	**0.9769**	**0.9633**	**0.9634**	**0.0062**
MGWAM	*w/o* MCA	0.9344	0.8853	0.8668	0.8551	0.9768	0.9611	0.9603	0.0066
	w/o splitWS	0.9342	0.8840	0.8672	0.8503	0.9745	0.9594	0.9599	0.0068
Baseline+DDAM		**0.9351**	**0.8858**	**0.8653**	**0.8559**	**0.9774**	**0.9584**	**0.9625**	**0.0070**
DDAM	*w/o* CEA	0.9343	0.8839	0.8642	0.8542	0.9754	0.9556	0.9611	0.0075
	w/o ASA	0.9333	0.8804	0.8633	0.8537	0.9737	0.9494	0.9576	0.0076
	w/o MB ASA	0.9344	0.8846	0.8627	0.8533	0.9770	0.9523	0.9592	0.0071

d) Effectiveness of DDAM: We use the same strategy to verify the effectiveness of DDAM. First, we independently verify the effectiveness of DDAM on the baseline, setting up Experiment 4. DDAM guides the aggregation features of the deepest two layers of the encoder, fully extracts global context information through dual-domain attention, and fuses it with the features of the first two layers at the element level. Finally, the same decoder is used to generate the final result. As shown in No. 4 of Table IV, it is clear that the model's performance has been improved after adding DDAM. Compared with it, the performance of various indicators of the F-measure has increased by approximately 0.61%–2.04%, the performance of various indicators of the E-measure has increased by around 0.67%–2.13%, MAE's performance has increased by 0.09%, and S_α has increased by 0.93%. Similarly, we verify the effectiveness of DDAM in the multimodule model, setting up Experiment 7. Compared with the experimental data of adding DDAM, the performance of $F_\beta^{\max}$ has increased by 0.07%, $E_\xi^{\max}$ has increased by 0.05%, MAE's performance has increased by 0.03%, and S_α has increased by 0.06%. As shown in Fig. 12, it presents a visual comparison of Experiment 7 and the complete model. It can be found that DDAM can improve the model's detection performance for small targets. In the visualization results, whether it is the attention to small targets or the generation of small-target masks, there are obvious improvements. All of these demonstrate that DDAM is effective.

In summary, we verify the effectiveness of each module through different ablation experiments, based on the baseline and the benchmarks of methods including all other modules. Based on the comparison of the above experimental data, all the modules we proposed are effective, and each module provides a solution to overcome the challenges in SOD.

2) Effectiveness of Each Component: To study the effectiveness of each component in the three modules, we set up multiple comparison experiments to separately verify the effectiveness of each component for the baseline and the overall network. To ignore the impact of other modules on the results and more accurately verify their effectiveness, we first set up ablation experiments based on a single module on the baseline. The experimental settings for verifying the

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TABLE VI
ABLATION RESULTS OF EVALUATING THE EFFECTIVENESS OF EACH COMPONENT IN LGIPNET. THE BEST ONE IN EACH COLUMN IS BOLD

Modules		EORSSD							
		$S_\alpha \uparrow$	$F_\beta^{max} \uparrow$	$F_\beta^{mean} \uparrow$	$F_\beta^{adp} \uparrow$	$E_\xi^{max} \uparrow$	$E_\xi^{mean} \uparrow$	$E_\xi^{adp} \uparrow$	$MAE \downarrow$
Full model		**0.9392**	**0.8924**	**0.8770**	**0.8623**	**0.9806**	**0.9753**	**0.9719**	**0.0053**
EAEM	w/o DNPB	0.9378	0.8919	0.8733	0.8534	0.9790	0.9723	0.9700	0.0058
MGWAM	w/o MCA	0.9389	0.8921	0.8756	0.8601	0.9804	0.9739	0.9696	0.0056
	w/o WS	0.9387	0.8911	0.8742	0.8564	0.9801	0.9711	0.9689	0.0058
DDAM	w/o CEA	0.9381	0.8903	0.8759	0.8602	0.9794	0.9686	0.9707	0.0057
	w/o ASA	0.9390	0.8918	0.8763	0.8610	0.9797	0.9699	0.9701	0.0055

effectiveness of each element in EAEM on the baseline are: 1) w/o split: remove the feature split and perform subsequent operations on a single branch; 2) w/o split SA: remove multibranch split and attention operation; 3) w/o DNPB: remove the DNPB in the module; the experimental settings for verifying the effectiveness of each element in MGWAM are: 4) w/o MCA: remove the multiscale attention part of this module, directly split the features, and perform subsequent operations; 5) w/o splitMSA: remove the feature split and weighted attention operation in this module; the experimental settings for verifying the effectiveness of each element in DDAM are: 6) w/o CEA: remove the CEA part of this module; 7) w/o ASA: remove the ASA part of this module; and 8) w/o MB ASA: remove the multibranch ASA operation in this module, only do single-branch ASA operation. The corresponding experimental results of the above experiments are shown in Table V. Second, we set up ablation experiments based on a single element on the overall network, the experimental data are shown in Table VI. The experiment includes all proposed modules except the verification element. The experimental settings for verifying the effectiveness of each element in the module on the overall network are: 1) w/o DNPB: remove the DNPB operation in the network; 2) w/o MCA: remove the multiscale attention operation in the network; 3) w/o WS: remove the weighted attention operation in the network; 4) w/o CEA: remove the CEA in the network; and 5) w/o ASA: remove the ASA in the network.

a) Ablation experiment inside the EAEM: To verify the effectiveness of each element in EAEM, we set up four experiments to validate its contribution to the baseline and the overall network. As shown in Table V, the ablation experimental data for the EAEM part reveals that all three components in EAEM improve the baseline. Although the split operation does not significantly improve, it can enhance the efficiency of the module's use of features without increasing the computational burden, enabling the attention mechanism and DNPB to extract more comprehensive feature information. Therefore, we can find that after removing the split and spatial attention operations, the performance of the model has significantly decreased, which validates the effectiveness of the split operation in positively guiding and combining spatial attention operations. When we remove DNPB, features from different channels cannot be differentially perceived through this module, thereby losing the ability to extract edge features. The data in Table V show that after removing DNPB, the performance of the model has greatly decreased, which validates that our multibranch differential perception strategy is feasible and effective. To further verify its contribution, we conducted ablation experiments on it in the whole model. The experimental data are shown in Table VI. The improvement of this module to the overall network is also quite obvious. This also verifies that EAEM is effective in improving the quality of target edge detection.

b) Ablation experiment inside the MGWAM: To verify the effectiveness of each element in MGWAM, we also set up four experiments to validate its contribution to the baseline and the overall network. MGWAM is the most important module in our network. We process the three layers of features generated by the backbone from high to low with this module. As shown in Table V, in the experiments based on the baseline, multiscale processing of features and channel attention operations, and weighted spatial attention operations after splitting, all significantly improve the performance of the baseline, which proves that these two components are effective. The experimental data corresponding to "Baseline + MGWAM" in Table V proves that when the two modules are applied to the baseline at the same time, the model achieves better performance. We also verify its effectiveness on the complete model, and the specific experimental data are shown in Table VI. When we sequentially remove these two modules in the whole model, the overall performance of the network has a significant performance decline. Therefore, the effectiveness of the MCA and split WS modules is verified through the above, and their combination efficiency is verified through various combinations.

c) Ablation experiment inside the DDAM: To verify the effectiveness of each element in DDAM, we use the same experimental settings for its validation. As shown in Table V and the DDAM part of Table VI, when we conduct ablation experiments on the internal structure of DDAM, the performance of the model will decrease to a certain extent, especially when we conduct ablation experiments on CEA and ASA, the performance of the model based on the baseline significantly decreases. Removing the CEA module, the model cannot fully mine deep semantic information, thus losing the acquisition of small-target information, while the ASA block can adaptively extract the context information of the target, making the detection results as complete as possible. According to the data in Table VI, the roles of the above two modules can be fully proven. We also verify the ASA module of multiple branches at the same time. It can be found that the performance of the entire module is improved by information interaction between multibranch features. We also verify its effectiveness on the complete model. The specific experimental data are shown in Table VI. When we remove these two modules from the whole model in turn, the overall performance of the network has a significant performance decline. It is worth mentioning that when we remove CEA from the whole network, the S_α index has a huge drop, and at the same time, other indicators also have different degrees of decline. This also shows that DDAM provides an effective approach to the challenge of difficult detection of small targets.

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D. Discussion

While LGIPNet demonstrates strong performance in enhancing long-range dependency modeling, small-object detection, and background suppression for ORSI-SOD, it still presents certain limitations. The inclusion of numerous attention computational modules results in high-computational complexity, potentially limiting its applicability in real-time scenarios. In addition, in environments characterized by low contrast and heavy noise interference, the edge extraction performance of the EAEM can be degrade, thus affecting the model's accuracy in identifying salient objects. In addition, when scenes contain both extremely large and extremely small objects, the ASA mechanism of the DDAM module may struggle to effectively handle such significant scale variations. The future research could focus on lightweight model design, multimodal feature fusion, and transfer learning strategies to enhance the model's accuracy and generalization capabilities.

V. Conclusion

In this article, we are among the few to propose an end-to-end ORSI-SOD solution based on Transformer and introduce the LGIPNet model. Specifically, we thoroughly analyze the features generated by the backbone at each level and design three modules: EAEM, MGWAM, and DDAM. EAEM is used for high-resolution shallow features, performing feature difference perception through splitting operations without increasing computational load, and enhancing the distinction between the edges and background of the final detection map. MGWAM achieves the local enhancement of multilevel features containing global information. MGWAM includes custom channel attention operations under multiscale feature changes, feature splitting operations, and weighted spatial attention operations. DDAM operates on the highest two levels of features, using CEA to guide the information interaction between cross-channel features, and then deeply mining the dependency information at different distances in high-level features through ASA, achieving effective interaction of deep semantic information. We validated our model through experiments on different datasets and comprehensive ablation experiments. The results validate the effectiveness of our three modules and the superiority of LGIPNet.

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